

Generation Interconnection System Impact Study Report

For Queue Project Q15-0441

Lost Canyon 100 kV

April 20, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0441. The Interconnection Customer proposes a 374.5 MW wind facility interconnecting at the Lost Canyon 100 kV substation in the Illinois area. The request was received on February 27, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 172 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$11,877,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Dayton'S Bluff-Lost Canyon 100 KV line	Mtn Island-Lost Canyon 100 KV	225.3	119	74.2
Mountain Island-Lost Canyon 100 KV	Dayton'S Bluff-Lost Canyon 100 KV	198.0	124	73.1
Mountain Island-Texon 100 KV	Lost Canyon-Dayton'S Bluff 100 KV	128.4	137	55.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dayton'S Bluff-Lost Canyon 100 KV line	Rebuild line with higher-capacity conductor	\$3,610,000	100% (\$3,610,000)
Mountain Island-Lost Canyon 100 KV	Rebuild line with higher-capacity conductor	\$7,031,000	100% (\$7,031,000)
Texon-Mtn Island 100 KV	Rebuild line with higher-capacity conductor	\$1,277,000	97% (\$1,236,000)
	Total allocated to project		\$11,877,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.